

SOLNETHNAYA, Russian FederationWMO: **305370**Lat: **54.030N**Lon: **108.255E**Elev: **464**StdP: **95.88**Time Zone: **8.00 (E08)**Period: **90-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -25.4	(d) -32.7	(e) 0.2	(f) -25.4	(g) -30.4	(h) 0.2	(i) -23.0	(j) 16.8	(k) -13.4	(l) 14.1	(m) -13.8	(n) 1.9	(o) 300	(p) 0.739

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.2	(c) 20.7	(d) 15.6	(e) 19.3	(f) 14.9	(g) 18.1	(h) 14.1	(i) 16.8	(j) 19.1	(k) 15.8	(l) 18.2	(m) 15.0	(n) 17.3	(o) 2.7	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.9	18.3	14.8	11.1	17.2	13.9	10.5	16.5	49.1	19.1	46.2	18.4	43.7	17.3	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	12.4	10.1	-29.5	25.3	2.8	2.0	-31.5	26.7	-33.2	27.8	-34.8	28.9	-36.8	30.4
			-29.5	18.4	2.9	1.4	-31.5	19.4	-33.3	20.2	-34.9	20.9	-37.0	21.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-16.6	-16.0	-9.9	-1.3	4.7	9.3	12.9	14.3	8.3	2.1	-5.5	-9.6	(6)
(7)		DBStd	11.25	5.70	5.47	5.72	4.14	3.12	2.59	3.17	2.53	2.96	3.88	5.34	4.44	(7)
(8)		HDD10.0	4118	826	729	618	339	164	44	8	3	66	247	466	608	(8)
(9)		HDD18.3	6888	1084	963	877	589	421	272	171	127	299	504	716	866	(9)
(10)		CDD10.0	275	0	0	0	0	1	22	98	137	17	1	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	0	2	2	0	0	0	0	(11)
(12)		CDH23.3	12	0	0	0	0	0	0	5	7	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	3.2	2.9	2.3	2.7	3.4	2.9	2.1	1.8	2.4	3.6	4.0	5.2	4.4	(14)
(15)	Precipitation	PrecAvg	401	17	8	9	19	30	46	72	67	43	22	34	34	(15)
(16)		PrecMax	536	60	16	30	37	55	98	154	112	85	49	53	59	(16)
(17)		PrecMin	263	5	2	2	8	9	14	21	25	15	5	15	19	(17)
(18)		PrecStd	58	11	3	6	8	12	21	33	24	18	10	9	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	0.5	6.1	11.6	15.6	19.5	23.5	23.7	17.4	13.7	7.2	2.1	(19)
(20)			MCWB	-3.5	-3.1	1.8	5.9	9.0	12.4	16.4	17.3	13.5	8.0	3.1	-1.0	(20)
(21)		2%	DB	-3.1	-2.8	2.9	8.4	13.2	17.2	21.3	20.5	15.9	10.4	5.0	-0.7	(21)
(22)			MCWB	-5.2	-5.3	-0.8	3.2	7.3	11.4	16.0	15.7	11.9	6.5	1.4	-3.3	(22)
(23)		5%	DB	-5.6	-5.5	0.6	6.3	11.5	15.5	19.7	19.3	14.6	9.0	3.6	-2.4	(23)
(24)			MCWB	-7.4	-7.3	-2.4	2.0	6.0	10.4	15.2	15.1	11.2	5.5	0.3	-4.8	(24)
(25)		10%	DB	-8.7	-8.0	-1.6	4.5	9.9	14.0	18.4	18.3	13.4	7.6	1.6	-3.8	(25)
(26)			MCWB	-10.2	-9.5	-4.0	0.8	5.0	9.4	14.5	14.4	10.0	4.2	-1.3	-5.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.3	-3.2	2.2	6.2	9.7	13.7	18.8	18.2	15.0	8.9	3.5	-0.7	(27)
(28)			MCDWB	-1.1	-0.1	5.7	11.4	14.9	17.0	21.9	21.9	17.2	12.2	6.5	1.5	(28)
(29)		2%	WB	-5.2	-5.3	-0.4	3.6	8.0	12.2	17.1	16.8	12.9	7.1	1.6	-3.1	(29)
(30)			MCDWB	-3.4	-3.2	2.7	7.8	12.4	16.0	19.5	19.1	15.1	10.1	4.5	-1.1	(30)
(31)		5%	WB	-7.5	-7.4	-2.3	2.1	6.5	11.1	15.9	16.1	11.6	5.9	0.4	-4.5	(31)
(32)			MCDWB	-5.7	-5.6	0.4	5.7	10.5	14.7	18.6	18.4	14.0	8.7	3.4	-2.6	(32)
(33)		10%	WB	-10.1	-9.4	-4.0	1.0	5.4	10.0	14.9	15.3	10.5	4.4	-1.0	-5.6	(33)
(34)			MCDWB	-8.8	-8.1	-1.5	4.4	9.3	13.3	17.8	17.4	12.9	7.2	1.6	-3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	7.8	7.3	7.8	7.9	7.5	6.2	6.0	6.3	5.7	5.3	(35)
(36)			MCDBR	9.0	9.9	10.0	10.1	10.6	10.2	10.1	7.2	7.3	7.9	7.5	6.2	(36)
(37)			MCWBR	7.8	8.0	7.3	7.0	6.6	6.3	5.9	4.8	5.7	5.9	6.1	5.4	(37)
(38)		5% WB	MCDWB	9.1	9.6	9.4	9.6	10.1	9.6	8.4	6.3	6.6	7.7	7.3	6.0	(38)
(39)			MCWBR	7.9	7.8	7.1	6.9	6.9	6.6	5.8	4.9	5.8	6.1	6.2	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.272	0.305	0.332	0.380	0.430	0.464	0.447	0.404	0.351	0.317	0.285	0.268	(40)	
(41)		taud	2.024	1.993	2.025	2.113	2.110	2.117	2.221	2.323	2.416	2.326	2.070	1.984	(41)	
(42)		Ebn at Noon	649	755	828	840	815	790	797	814	818	759	626	556	(42)	
(43)		Edh at Noon	73	107	130	137	146	147	131	111	89	77	69	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.98	1.98	3.40	4.59	5.48	6.03	5.47	4.59	3.22	1.84	0.92	0.65	(44)	
(45)		RadStd	0.08	0.12	0.17	0.33	0.38	0.38	0.47	0.21	0.25	0.19	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	-1.55	-2.01	N/A	N/A	N/A	N/A	N/A	+44	N/A

Nomenclature: See separate page